

Estimation of under-reporting influenza cases in Hong Kong based on bayesian hierarchical framework

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ABSTRACT

Influenza remains a global challenge, imposing a significant burden on society and the economy. Many influenza cases are asymptomatic, leading to greater uncertainty and the under-reporting of cases in influenza transmission and preventing authorities from taking effective control measures. In this study, we propose a Bayesian hierarchical approach to model and correct under-reporting of influenza cases in Hong Kong, incorporating a discrete-time stochastic, Susceptible-Infected-Recovered-Susceptible (DT-SIRS) model that allows transmission rate to vary over time. The incidence of influenza exhibits seasonality. To examine the relationship between meteorological factors and seasonal influenza activity in subtropical areas, five meteorological factors are included in the model. The proposed model explores the effects of meteorological factors on transmission rates and disease detection covariates on under-reporting, and the inclusion of the DT-SIRS model enables more accurate inference regarding true disease counts. The results demonstrate that under-reporting rates of influenza cases vary significantly in different years and epidemic seasons. In conclusion, our method effectively captures the dynamic behavior of the disease, and we can accurately estimate under-reporting and provide new possibilities for early warning of influenza based on meteorological data and routine surveillance data.

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1. Introduction

Influenza is an acute respiratory infection caused by the influenza virus. The risk of respiratory diseases is increasingly, and influenza has become a significant public health problem threatening human health. Seasonal influenza causes approximately 3–5 million serious illness and 290,000 to 650,000 deaths worldwide annually (World Health Organization, 2018). Investigating the issue of under-reporting in disease surveillance and reporting is essential to accurately and timely evaluate disease burden (Kupek, 2021). Under-reporting of surveillance data can lead to biased statistical conclusions that impact parameter estimation and forecasting, thereby under-estimating the risk of disease. The COVID-19 pandemic has prompted us to reconsider the role of asymptomatic infections in influenza transmission (Montgomery et al., 2024). About 30–50 % of influenza cases are asymptomatic (Cohen et al., 2021; Lau et al., 2010). These undetected individuals could potentially drive the influenza pandemic (Fielding et al., 2015). Therefore, recognizing the severity of under-reporting in a timely manner will

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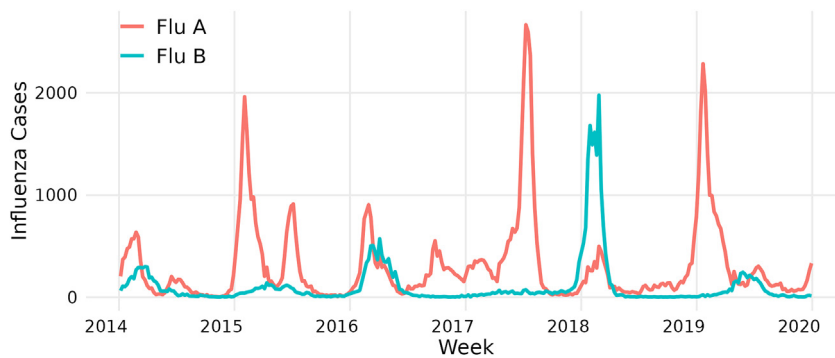


Fig. 1. Number of weekly cases of Flu-A and Flu-B in Hong Kong from week 1, 2014 to week 52, 2019.

help us more accurately assess the true extent of the influenza epidemic, thereby aiding governments in formulating more effective prevention and control strategies to protect public health.

There are relatively few approaches available to account for under-reporting in count data, among which classic methods include the censored likelihood method (Bailey et al., 2005; de Oliveira et al., 2017) and the hierarchical count framework (Lope & Demirhan, 2022; Winkelmann & Zimmermann, 1993). These models typically rely on regression-based techniques, which make it difficult to dynamically capture the transmission of infectious diseases. As a result, they are more suitable for large-scale populations—such as modeling at the nation level—while their performance tends to fall short when applied to smaller regions or individual cities, where more precise estimations are required. The Poisson-logistic model proposed by Winkelmann et al., implemented within the Bayesian framework consisting of an incidence component model and a case detection probability component model, has been widely applied across various fields. In this paper, we extended the Poisson-logistic model and applied it to influenza data. In particular, since the model cannot directly account for the dynamic spread of disease in populations, we replaced the function expression in the incidence component with the model proposed by Lawson and Song (2010) and developed a discrete-time compartmental model to simulate the transmission of seasonal influenza in the population. The transmission rate is allowed to vary over time, and considers various forms of period effects (Bucyibaruta et al., 2023) to capture variations in influenza activity between different seasons. As we assume time and state variables are discrete, this compartmental model is a discrete-time, stochastic, Susceptible-Infectious-Recovered-Susceptible (DT-SIRS) model. The Poisson-logistic model is integrated with the DT-SIRS model through an extended incidence component, creating a more flexible new model. This new model can simultaneously derive predictive distributions for both true counts and true transmission rates by incorporating covariates that impact disease detection and transmission, thereby quantifying the uncertainty in under-reporting and transmission.

There is growing interest in the correlation between climate variables and seasonal influenza activity. The seasonal characteristics of influenza are more diverse in subtropical areas (Liu et al., 2018). Hong Kong, which belongs to a humid subtropical climate, is one of the areas that experience highly irregular influenza outbreaks every year (Fig. 1). As an attractive city on the southern coast of China, Hong Kong attracts large numbers of tourists from around the world each year, and the influx of these travelers could increase the importation of influenza cases and further promote local transmission. Due to these unique climatic and socio-economic characteristics, influenza outbreaks may occur one or more times in a year in Hong Kong. Previous researches have explored the relationship between meteorological factors and the transmission rate of influenza, finding that simulation from transmission rates incorporating meteorological factors perform much better than those without meteorological factors (He et al., 2013). However, most studies have focused on tropical or temperate regions, with little knowledge about the relationship between various climate variables and influenza in subtropical areas. To supplement the studies on seasonal influenza in subtropical areas, we included meteorological factors (mean temperature (MeanT), relative humidity (RH), diurnal temperature range (DTR), wind velocity (Wv), rainfall) in the transmission function, thus elucidating in detail the associations of multiple climate variables with influenza subtype outbreaks in Hong Kong.

The article is structured as follows: In Section 2, we provide a detailed introduction to the data used and review the Poisson-logistic model of Winkelmann et al. We then discuss extensions to the model, including the DT-SIRS model and various forms of period effects. In Section 3, we present the application to Hong Kong influenza data and some simulation experiments aimed at demonstrating the superiority of the extended model. Finally, in Section 4, we conclude the paper with a general discussion of the proposed model and offer suggestions for future research in this field.

2. Methods

2.1. Data source

The Hong Kong Centre for Health Protection publishes weekly network reports compiled by sentinel private medical practitioner (PMP) clinics and sentinel general outpatient clinics (GOPC) on the proportion of patients seeking treatment for

influenza-like illness (ILI), defined as fever ($\geq 38^{\circ}\text{C}$) along with sore throat or cough (Centre for Health Protection/Department of Health of the Government of the Hong Kong Special Administrative Region, 2023). In this study, we used sentinel surveillance data from PMP and GOPC covering the period from week 1 of 2014 to week 52 of 2019. The Centre for Health Protection conducts testing on respiratory samples provided by the Public Health Laboratory Services Branch and the Hospital Authority for monitoring and diagnostic purposes (Centre for Health Protection). We obtained data on the weekly number of positive samples for influenza A (Flu-A) and influenza B (Flu-B), as well as the weekly number of samples tested. All data are publicly available from the official website of the Hong Kong Centre for Health Protection and are provided in weekly format, requiring no additional preprocessing.

Meteorological variables including daily mean temperature, RH, Wv, and rainfall were obtained from the Hong Kong Observatory (Hong Kong Observatory). The DTR is the difference between the daily maximum temperature and the daily minimum temperature. The daily meteorological factors are averaged during the influenza surveillance weeks.

2.2. Distributed lag nonlinear models (DLNMs)

The influence of meteorological factors on influenza cases is sometimes not limited to the current point in time, but can involve time delays. To better assess the potential nonlinear and lagged effects of climate features on influenza transmission, DLNMs were constructed separately for Flu-A and Flu-B (Gasparrini, 2011). Spearman correlation analysis was used to avoid multicollinearity among climate variables (Table S1). We constructed a Poisson regression with a quasi-Poisson function that allows for over-dispersion of weekly influenza cases. The model is formulated as follows:

$$\log[E(z_t)] = \alpha + cb(\text{climate variables}, \text{lag}, df) + \sum ns(X_j, df) + ns(\text{time}, df \times 6) + \text{factor}(\text{holiday}) \quad (1)$$

where $E(z_t)$ is the expected weekly numbers of Flu-A or Flu-B cases in week t ; α is the intercept; $cb(\cdot)$ represents the cross-basis matrix of the meteorological factor, and $ns(\cdot)$ is a cubic spline function. X_j refer to other meteorological variables excluding the meteorological factor in cross-basis matrix. *Time* represents seasonality and long-term trends in influenza, controlled using a cubic spline function with 5 degrees of freedom (df) per year for Flu-A and 3 df per year for Flu-B. *Holiday* is a binary variable to adjust for the effects of public holidays.

We established DLNMs for MeanT, RH, DTR, Wv and rainfall, respectively. The maximum temporal lag was set at 3 weeks, which was based on the incubation period of influenza and potential lagged effects by previous researches (Dai et al., 2018). Based on the principle of minimizing the Quasi-Akaike Information Criterion (QAIC), the df for the cross-basis of various meteorological factors were determined (Table S2). We then calculated the relative risk (RR) with the corresponding 95 % confidence interval (CI) in Table 1. The RR, which represents the incidence risk resulting from a unit change in a specific meteorological variable relative to its reference value, is used to quantify the effects of climatic conditions on influenza.

2.3. Poisson-logistic model

The Poisson-logistic model proposed by Winkelmann et al. is implemented through a Bayesian hierarchical framework. The model consists of a binomial component for the observed counts z , conditional on the underlying unobserved true counts y , which are assumed to follow a Poisson distribution with mean λ . The basic form of the model is as follows:

Table 1
The RRs with 95 % CI of Flu-A and Flu-B associated with climate variables at different lags in Hong Kong from week 1, 2014 to week 52, 2019.

		RR			
		0-week lag	1-week lag	2-week lag	3-week lag
MeanT					
Flu-A	10.3 °C vs 25.0 °C	1.04(0.53–2.03)	1.46(0.91–2.36)	1.65(1.05–2.60) ^a	1.62(1.01–2.60) ^a
Flu-B	10.3 °C vs 25.0 °C	2.44(1.15–5.19) ^a	2.28(1.33–3.90) ^a	3.20(2.04–5.01) ^a	3.41(2.13–5.46) ^a
RH					
Flu-A	94 % vs 80 %	1.73(1.02–2.93) ^a	0.80(0.53–1.20)	1.36(0.92–2.01)	1.40(0.95–2.06)
Flu-B	94 % vs 80 %	3.40(1.84–6.30) ^a	1.57(0.96–2.56)	1.88(1.17–3.01) ^a	1.78(1.08–2.94) ^a
DTR					
Flu-A	6.2 °C vs 4.5 °C	1.58(1.17–2.15) ^a	1.26(0.95–1.66)	0.97(0.73–1.28)	0.84(0.64–1.08)
Flu-B	6.2 °C vs 4.5 °C	1.30(1.03–1.64) ^a	1.18(1.00–1.39) ^a	0.86(0.70–1.06)	0.87(0.71–1.05)
Wv					
Flu-A	4.4 m/s vs 6.3 m/s	1.17(0.95–1.45)	1.30(1.07–1.57) ^a	1.27(1.04–1.54) ^a	1.19(0.98–1.45)
Flu-B	4.4 m/s vs 3.2 m/s	1.58(0.94–2.65)	0.98(0.62–1.55)	1.86(1.02–3.39) ^a	0.96(0.56–1.66)
Rainfall					
Flu-A	15.4 mm vs 56.2 mm	1.13(0.87–1.21)	1.22(0.95–1.15)	1.31(1.02–1.24) ^a	1.20(0.94–1.24)
Flu-B	15.4 mm vs 56.2 mm	1.57(0.62–4.00)	2.91(1.03–8.18) ^a	1.71(0.56–5.16)	0.81(0.36–1.80)

^a Significant results.

$$z_t | y_t \sim \text{Binomial}(y_t, \pi_t) \quad (2)$$

$$\log\left(\frac{\pi_t}{1 - \pi_t}\right) = b_0 + \sum_{k=1}^K b_k v_t^k \quad (3)$$

$$y_t \sim \text{Poisson}(\lambda_t) \quad (4)$$

$$\log(\lambda_t) = a_0 + \sum_{j=1}^J a_j x_t^j \quad (5)$$

The model was assumed that observed data z are under-reported, where π_t denotes the probability of detecting influenza cases at time t , and $\mathbf{V} = \{v_t^k\}$ represents the vector of covariates associated with the disease detection process. This allows quantification of the severity of under-reporting and the impact of associated covariates. λ_t is the true disease incidence and $\mathbf{X} = \{x_t^j\}$ is the vector of covariates that induces the number of disease cases with the offsets such as population counts. By using integration and Bayes' rule (Stoner et al., 2019):

$$z_t \sim \text{Poisson}(\pi_t \lambda_t) \quad (6)$$

Thus, efficient sampling can be carried out using Markov chain Monte Carlo (MCMC).

2.4. DT-SIRS model

In Eq. (5), the incidence component λ_t is derived from covariates including population counts, potentially assumed that all individuals are susceptible without considering the dynamic spread of influenza within the population. To address this issue, we have developed a discrete-time Bayesian stochastic SIRS model focused on describing seasonal influenza dynamics. Generally, individuals remain immune after recovering from an infection but after some period of time they may become susceptible again due to the rapidly evolving nature of the influenza virus (Yaari et al., 2013). Therefore, for diseases such as influenza that confer temporary immunity but have become endemic, it is necessary to consider the process by which individuals lose their immunity and revert to a susceptible state.

We proposed a model that captures the dynamic changes in infectious diseases over time by incorporating the number of cases from the previous period as a potential explanatory variable for disease incidence. Let S_t , I_t , and R_t denote the number of susceptible, infectious, and recovered individuals in week t ($t = 1, \dots, n$) respectively. The number of newly infected cases y_t in week t is assumed to follow a Poisson distribution, with the expected value given by a product of susceptible individuals S_{t-1} and infected cases I_{t-1} in week $t - 1$. Therefore, $y_t \sim \text{Poisson}(\lambda_t)$, and λ_t is given by (Lawson & Song, 2010):

$$\lambda_t = S_{t-1} \times I_{t-1} \times \beta_t \quad (7)$$

$$\log(\lambda_t) = \log(S_{t-1}) + \log(I_{t-1}) + \log(\beta_t) \quad (8)$$

$$\log(\beta_t) = a_0 + \sum_{j=1}^J a_j x_t^j + r_t \quad (9)$$

Here, β_t refers to the true weekly transmission rate for week t ; a_0 represents the intercept, indicating the log-transformed baseline transmission rate during the study period; $(a_1, \dots, a_J)$ is the vector of the regression coefficient for the covariates $\mathbf{X} = \{x_t^j\}$; r_t is used to capture unobserved seasonal fluctuations, which will be discussed in more detail later.

In the discrete-time model, the number of individuals in each compartment is examined in a discrete time step. The number of susceptible S_t , infectious I_t , and recovered R_t individuals at week t are updated through the following recursion equations:

$$S_t = S_{t-1} - y_t + \eta R_{t-1}$$

$$I_t = I_{t-1} + y_t - \gamma I_{t-1} \quad (10)$$

$$R_t = R_{t-1} + \gamma I_{t-1} - \eta R_{t-1}$$

where η is the proportion of recovered individuals who lose their immunity and become susceptible again per unit of time (weeks); γ represents the proportion of infectious individuals who recover per unit of time (weeks). By substituting Eq. (8) for Eq. (5), we introduced the DT-SIRS model, which enables iterative updating of the unobserved true counts y within the compartment model.

To capture the seasonal effects, we explored different models ranging from Model 1 to Model 3 based on a variety of expressions for r_t . In this article, we used **Model 1** (M_1) as a benchmark for comparison and considered two forms of r_t . The first one adopts a fixed weekly seasonal effects, while the second expression uses sine-cosine curves (Corberán-Vallet & Santonja, 2014).

In **Model 1a** (M_{1a}), r_t varies over time with a period of 1-year (52 weeks) where each weekly effect ϕ_p , remains constant over years, so there are 52 fixed values $\phi_1, \phi_2, \dots, \phi_{52}$. The specific expression is as follows:

$$r_t = \phi_{p(t)} \quad (11)$$

where

$$p(t) = \begin{cases} w(t), & \text{if } w(t) \neq 0 \\ 52, & \text{otherwise} \end{cases}, w(t) = t \bmod 52 \quad (12)$$

In **Model 1b** (M_{1b}), r_t is considered as a sinusoidal curve to effectively reduce the number of parameters:

$$r_t = \sum_{h=1}^H \left(m_h \cos\left(\frac{2h\pi t}{52}\right) + n_h \sin\left(\frac{2h\pi t}{52}\right) \right) \quad (13)$$

where H represents the number of sine-cosine waves selected, and the value of H will be discussed further in Section 3.1.

In **Model 2** (M_2), r_t is modeled using fixed 4-week period effects, resulting in 13 fixed values $\phi_1, \phi_2, \dots, \phi_{13}$ (Bucyibaruta et al., 2023):

$$r_t = \phi_{q(t)} \quad (14)$$

where $q(t)$ is given by:

$$q(t) = \begin{cases} g(t) \bmod 13 \\ 13, & \text{if } g(t) \bmod 13 = 0 \end{cases} \quad (15)$$

with

$$g(t) = \begin{cases} \text{floor}\left(\frac{t}{4}\right) + 1, & \text{if } \text{mod}\left(\frac{t}{4}\right) \neq 0 \\ \text{floor}\left(\frac{t}{4}\right), & \text{if } \text{mod}\left(\frac{t}{4}\right) = 0 \end{cases}$$

Model 3 (M_3) defines r_t with different 4-week period effects over years:

$$r_t = \phi_{g(t)} \quad (16)$$

where

$$g(t) = \begin{cases} \text{floor}\left(\frac{t}{4}\right) + 1, & \text{if } \text{mod}\left(\frac{t}{4}\right) \neq 0 \\ \text{floor}\left(\frac{t}{4}\right), & \text{if } \text{mod}\left(\frac{t}{4}\right) = 0 \end{cases} \quad (17)$$

The model flowchart is shown in Fig. S1.

3. Model application

3.1. Parameter setting

Applying the Bayesian framework of this study using the Hong Kong Flu-A and Flu-B dataset, the true incidence, case detection probability and true transmission rate are estimated through Eqs. (2)–(4), (6)–(10). Let y_t and z_t respectively represent the true and recorded counts of influenza cases in Hong Kong in week $t \in \{1, 2, \dots, 313\}$ (from 2014 to 2019). Fig. 1 reveals that recorded influenza cases of Flu-A and Flu-B show a clear seasonal effect, with Flu-A peaking in early summer and winter, while Flu-B is mainly detected between March and May. Some of this variability may be attributed to seasonal covariates that influence influenza transmission. To allow for this, meteorological factors were considered as covariates in Eq (9), where $x_t^{(1)} = \text{MeanT}$, $x_t^{(2)} = \text{RH}$, $x_t^{(3)} = \text{DTR}$, $x_t^{(4)} = \text{Wv}$, $x_t^{(5)} = \text{Rainfall}$.

$$\log(\beta_t) = a_0 + f_1(x_{t-2}^{(1)}) + f_2(x_t^{(2)}) + f_3(x_t^{(3)}) + f_4(x_{t-2}^{(4)}) + f_5(x_{t-1}^{(5)}) + r_t \quad (18)$$

$f_1(\cdot), f_2(\cdot), \dots, f_5(\cdot)$ represent orthogonal polynomial functions of degrees 2, 2, 2, 3 and 2, respectively, aiming to reflect the nonlinear relationship between meteorological factors and influenza (Figs. S3–S4). Based on DLNMs, we analyzed the lagged associations between meteorological variables at specific values and influenza by different time lags, relative to the reference values (Table 1). Overall, low MeanT, moderate Wv, and moderate rainfall promote the spread of both influenza subtypes with sustained effects, while high RH and large DTR show immediate effects on influenza transmission, with the highest RR value at a lag of 0-week occurs. In Eq. (18), $x_{t-2}^{(1)}, x_t^{(2)}, x_t^{(3)}, x_{t-2}^{(4)}, x_{t-1}^{(5)}$ were selected with lag times of 2, 0, 0, 2, and 1 week, respectively, with these values determined based on the maximum lag weeks of the RRs in Table 1.

When characterizing the under-reporting mechanism, we considered the covariates $v_t^{(1)}$ = tested (the number of respiratory samples tested), $v_t^{(2)}$ = ILL_PMP (the average consultation rate for ILI among sentinel private medical practitioner clinics), $v_t^{(3)}$ = ILL_GOPC (the average consultation rate for ILI among sentinel general outpatient clinics) for the case detection probability. The first covariate $v_t^{(1)}$ reflects the scope and intensity of influenza detection, while the latter two embody the detection and reporting of influenza cases in medical facilities. Table 2 shows the summary statistics of the above covariates and influenza cases, while the distribution of covariates can be seen in Fig. S2. Then the case detection probability in Eq. (3) can be expressed in the following:

$$\log\left(\frac{\pi_t}{1 - \pi_t}\right) = b_0 + b_1 \times v_t^{(1)} + b_2 \times v_t^{(2)} + b_3 \times v_t^{(3)} \quad (19)$$

According to the different expressions of r_t , four models were developed separately for each of Flu-A and Flu-B: Models $M_{A-1a}, M_{A-1b}, M_{A-2}$ and M_{A-3} for Flu-A, and models $M_{B-1a}, M_{B-1b}, M_{B-2}$ and M_{B-3} for Flu-B.

We have completed the model construction and will next proceed with prior settings and parameter estimation. First, we assumed that $S_1 = 65\%$ of the population size in the first study week. The average recovery time from influenza is 7 days and the average duration of immunity loss is 365 days (Kim et al., 2017; Wang et al., 2021). So, we could set $\gamma = 1$ (week) and $\eta = 1/52$ (week). Second, we assumed diffuse normal priors for all regression parameters of the covariates in Eqs. (18) and (19), $a_j \sim N(0, 10^2), j = 0, \dots, 12, b_k \sim N(0, 10^2), k = 0, 1, 2, 3$. In models M_{1a}, M_2 and M_3 , the parameters in r_t were assumed to follow Gaussian distributions with mean zero, $\phi_{p(t)} \sim N(0, \epsilon_p^2), \phi_{q(t)} \sim N(0, \epsilon_q^2), \phi_{g(t)} \sim N(0, \epsilon_g^2)$; For model M_{1b} , H represents the number of sine-cosine waves selected. Too few waves may fail to fit the curve sufficiently, while too many may result in an excessive number of parameters and overfitting. Based on experience and fitting results, we set $H = 2$ (Domenech De Cellès et al., 2019), with priors for m_h and n_h ($h = 1, 2$) in Eq. (13) specified as $N(0, \epsilon_m^2)$ and $N(0, \epsilon_n^2)$, respectively. Finally, the priors for the variance parameters $\epsilon_p, \epsilon_q, \epsilon_g, \epsilon_m$ and ϵ_n were specified as zero-truncated $N(0, 1)$.

3.2. Simulation study

We conducted simulation studies to evaluate the performance of the proposed models. Consider the following model:

$$z_t | y_t \sim \text{Binomial}(y_t, \pi_t) \quad (20)$$

$$\log\left(\frac{\pi_t}{1 - \pi_t}\right) = b_0 + b_1 v_1 \quad (21)$$

$$y_t \sim \text{Poisson}(\lambda_t) \quad (22)$$

Table 2
Statistics of covariates and influenza cases in Hong Kong, 2014–2019.

	Mean	Min	Median	Max	SD
Flu-A	308	3	154	2668	425.4
Flu-B	106	0	34	1978	243.6
MeanT (°C)	23.9	10.3	25.0	30.7	5.0
RH (%)	78.5	49.0	80.0	94.0	7.5
DTR (°C)	4.4	2.2	4.5	7.0	0.9
Wv (m/s)	4.6	2.8	4.6	7.4	0.8
Rainfall (mm)	6.7	0.0	1.7	70.5	11.0
Tested	3958	1486	3987	9638	1346.5
ILL_PMP	42.9	10.0	42.4	82.3	11.8
ILL_GOPC	4.9	1.4	4.4	14.2	2.3

$$\lambda_t = S_{t-1} \times I_{t-1} \times \beta_t \quad (23)$$

$$\log(\beta_t) = a_0 + a_1 x_1 + r_t \quad (24)$$

with $b_0 = 0$, $b_1 = 2$, $a_1 = -1$ and $\varepsilon_p = \varepsilon_q = \varepsilon_g = \varepsilon_m = \varepsilon_n = 0.5$. A total of 156 data points were simulated and the covariates v_1 and x_1 were sampled from $Unif(-1, 1)$. One goal of this simulation is to examine the sensitivity of the model to the data generation mechanism. To achieve this, susceptible individuals S_t and infected individuals I_t were generated based on an Agent-based model (ABM) and implemented by NetLogo software. The agents in the model represent individuals, each agent was defined in three states: susceptible, infectious, and recovered. Additionally, each agent has specific characteristics (such as infectious period and immunity waning period) and can be in any of the different states. The initial state of the agents is set to “susceptible”, with some agents set to “infectious” to serve as sources of transmission. By simulating individual behaviors and their interactions while considering the heterogeneity among individuals, ABM can effectively reveal the dynamic characteristics of infectious disease transmission. We ran separate simulations for population sizes of 500, 2,000, and 20,000 individuals. All parameters’ priors were set to the same as in the influenza models.

An important indicator for evaluating the performance of the model is the proportion of reporting probabilities within the corresponding 95 % posterior predictive intervals (PIs), known as the coverage. Figs. S5–S7 show the coverage of the four models in three scenarios. We found that model M_3 performed quite well and the fitted values closely overlap the observed curves in all three scenarios. However, the performance of models M_{1a} , M_{1b} and M_2 varies slightly. When applied to simulated data with population sizes of 500 and 2,000, M_{1b} and M_2 either underestimate or overestimate the reporting probabilities, while the performance of M_{1a} is similar to that of M_3 . In the case of population size of 20,000, models M_{1b} , M_2 and M_3 outperform M_{1a} . Not only the coverage of four models has reached 99 %, but the 95 %PIs of the models gradually decrease as the population size increases, leading to more precise conclusions (Figs. S5–S7).

3.3. Model checking and model selection

All models were implemented using the “nimble” package in R software (de Valpine et al., 2017). Two chains were used, each producing 2000 samples following 100,000 iterations, with 50,000 burn-in periods and thinning by 25. Convergence of MCMC chains was assessed using the Gelman-Rubin statistics with “coda” package in R.

Model performance is evaluated by the deviance information criterion (DIC) and the root mean square error (RMSE). Models with lower DIC values are preferred as they achieve a more optimal combination of fit and parsimony (Table 3). RMSE is the comparison of the data point and the posterior mean estimate. A comparison of the DIC for models M_{A-1a} and M_{A-2} reveals that the two models are close in terms of overall fit, both superior to model M_{A-1b} but significantly inferior to model M_{A-3} . In addition, the comparison of RMSE indicates that model M_{A-3} is better than the other three models. Likewise, the model M_{B-3} is also the optimal model. Fig. 2 shows that models M_{A-3} and M_{B-3} , which represent the model with r_t modeled using different 4-week period effects over years, fit the data well.

Furthermore, we visualize the empirical cumulative distribution functions (ECDF) of the observed counts and posterior predictive samples for M_{A-1b} , M_{A-2} , M_{A-3} and M_{B-1b} , M_{B-2} , M_{B-3} in Fig. 3, in which the ECDFs are plotted on the log scale for better visualization. The dark green (black) solid lines in each subplot represent the ECDF of observed counts, while the yellow (brown) lines are ECDFs of posterior predictive samples provided by each model. In the first half, there is a discrepancy between the observed ECDFs and the predicted ECDFs in (a), (b) and (d), (e). Conversely, subplots (c) and (f) illustrate a nearly overlapping curve between the observed ECDFs and the predicted ECDFs. Similar to the DIC and RMSE results, the check based on the posterior predictive indicates that M_{A-3} and M_{B-3} exhibit the best fit.

3.4. Results

The effects of 2-week lagged MeanT on β_t of Flu-A and Flu-B are shown in the top-left and bottom-left panels of Fig. 4, indicating a negative correlation between MeanT and influenza incidence. Flu-B infection tends to peak in winter and spring, this is because lower temperatures can significantly increase the risk of contracting influenza. This effect is approximately 0.6 on the log scale, indicating that the transmission rate of Flu-B in Hong Kong is 1.8 times higher at low MeanT (<15 °C) than at

Table 3
DIC and RMSE for different models.

	M_{A-1a}	M_{A-1b}	M_{A-2}	M_{A-3}
DIC	135058	148233	142393	3276
RMSE	418.15	447.61	437.45	43.43
	M_{B-1a}	M_{B-1b}	M_{B-2}	M_{B-3}
DIC	17192	22295	19197	2156
RMSE	89.68	145.98	111.62	16.46

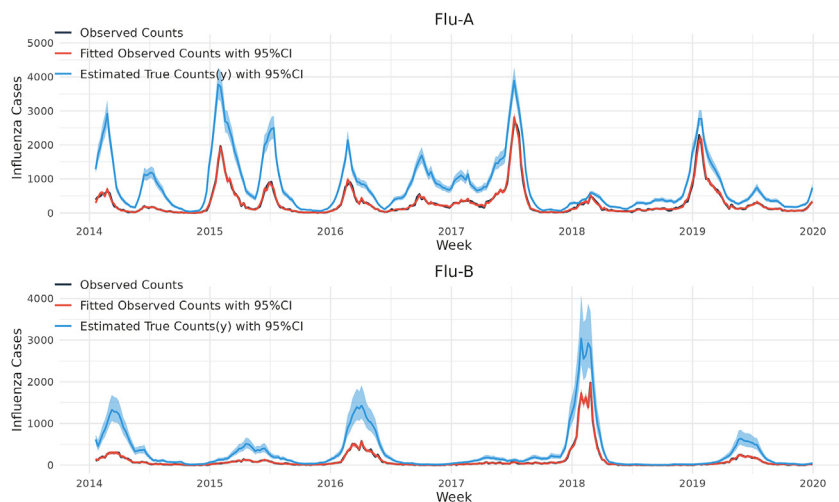


Fig. 2. Reported data of Flu-A and Flu-B (black), fitted observed values (red) and corresponding 95 % credible intervals (CI), and estimated true counts and corresponding 95 % CI (blue) of Hong Kong. The fitted observed counts and true counts with their 95 % CI are obtained from the posterior prediction samples estimated by the models $M_{A,3}$ and $M_{B,3}$, where the r_t is modeled using a different 4-week period effect over years.

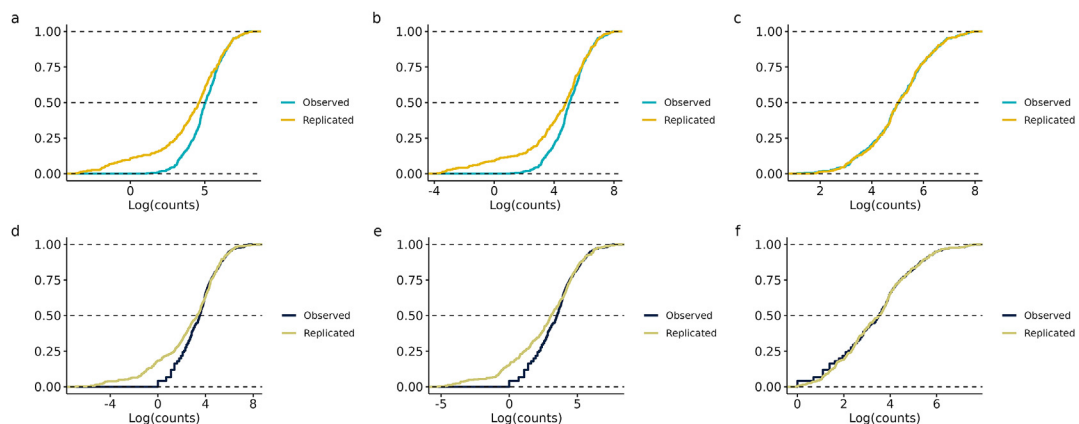


Fig. 3. Comparison between the empirical distribution of the logarithmic observed Flu-A and Flu-B counts and the distribution of replicated samples from the posterior predictive distribution based on $M_{A,1b}$, $M_{A,2}$, $M_{A,3}$, $M_{B,1b}$, $M_{B,2}$ and $M_{B,3}$. (a), (d): represent models $M_{A,1b}$ and $M_{B,1b}$ with sinusoidal curve; (b), (e): represent models $M_{A,2}$ and $M_{B,2}$ with same 4-week period effect over years; (c), (f): represent models $M_{A,3}$ and $M_{B,3}$ with different 4-week period effect over years.

high MeanT ($>25^{\circ}\text{C}$). Flu-A occurs primarily in winter and summer, suggesting that both low and high temperatures can promote its outbreak. The range of this effect is approximately 0.3, implying that the impact of low MeanT is stronger than high MeanT. The upper and lower right graphs in Fig. 4 show the effect of RH on β_t , revealing markedly different trends. RH is strongly positively correlated with Flu-A and negatively correlated with Flu-B. This means that wet weather promotes the spread of Flu-A, while dry conditions are more conducive to the transmission of Flu-B.

The left column of Fig. 5 shows the effect of DTR on β_t . The polynomial decreases monotonically for most of the range covered by the data ($\text{DTR} < 6^{\circ}\text{C}$), before increasing for higher values. This suggests that both small DTR and large DTR promote influenza transmission. The effect of 2-week lagged Wv is less pronounced: Wv is negatively correlated with Flu-A and positively associated with Flu-B. Finally, the right column of Fig. 5 shows the effect of 1-week lagged rainfall. Rainfall is positively associated with Flu-A, with high rainfall posting the risk of influenza. Meanwhile, the Flu-B curve shows a trend of an initial increase followed by a decrease, indicating that moderate rainfall (10–20 mm) may increase the risk of influenza.

The Hong Kong Centre for Health Protection publishes a weekly influenza activity report, which assesses influenza prevalence by setting the baseline threshold of positivity for seasonal influenza virus detection as the average weekly influenza positivity rate plus 1.96 standard deviations during the preceding four year's non-seasonal periods. If the positivity rate exceeds this baseline threshold, influenza is considered to have entered the epidemic season, as shown by the grey shaded area in Fig. 6. Overall, there are two peaks of influenza activity in Hong Kong each year. The orange curve in Fig. 6

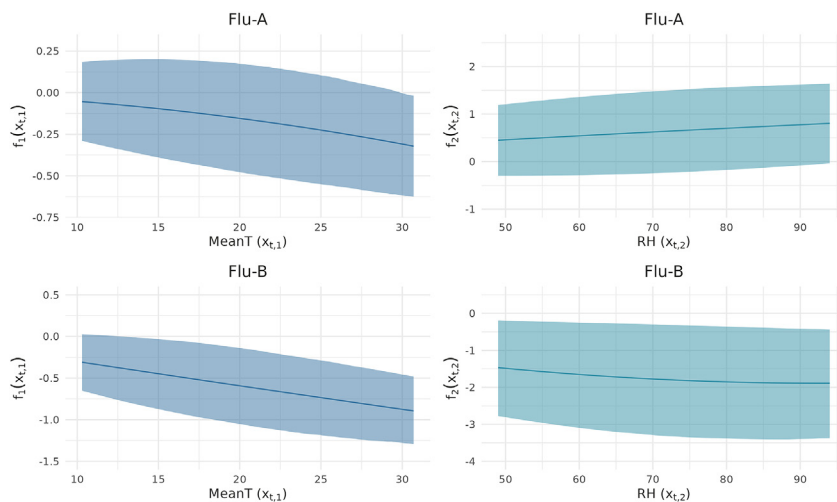


Fig. 4. Posterior mean predictions of the effects of MeanT and RH on the transmission rate of Flu-A and Flu-B, with associated 95 % CI.

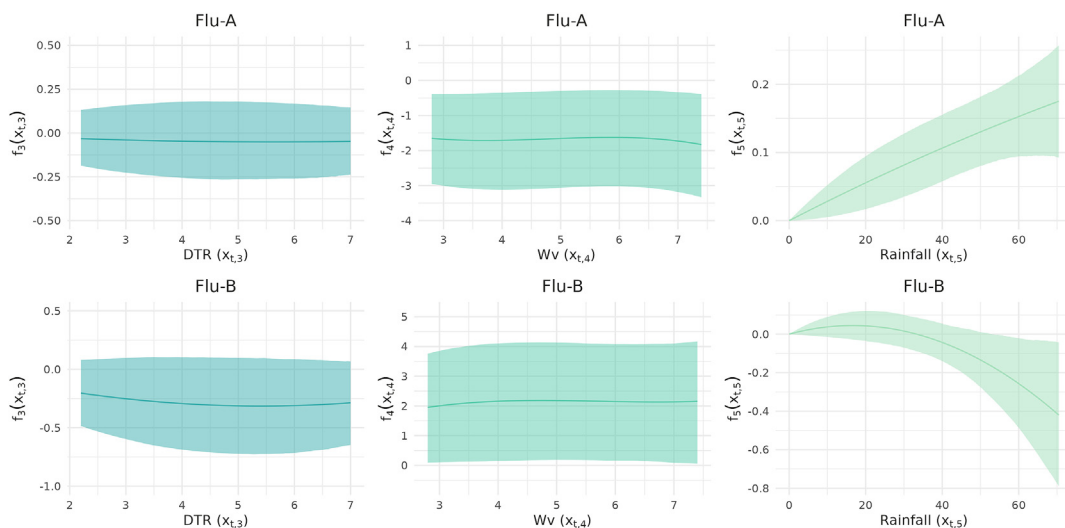


Fig. 5. Posterior mean predictions of the effects of DTR, Wv and rainfall on the transmission rate of Flu-A and Flu-B, with associated 95 % CI.

shows the relationship between reported influenza cases and the reporting probability π_t . It is obvious that π_t increases significantly during each epidemic season compared to non-epidemic periods and shows a trend of annual growth.

Table 4 shows the summary statistics of reporting probability π_t for Flu-A and Flu-B in different years (see Fig. S8 for weekly results). From 16.6 % in 2014 to 46.3 % in 2019, the annual mean of π_t for Flu-A has continuously increased, while Flu-B has risen from 19.8 % in 2014 to 37.8 % in 2019 with an annual average growth rate of 13.8 %, lower than that of 22.8 % for Flu-A. Table 4 also provides statistics on π_t on influenza during each annual epidemic season (choosing the longest epidemic season of each year for comparison). During epidemic seasons, the reporting probability π_t tends to be higher than the annual average, reflecting the additional investment of government resources in public health and heightened awareness of public protection. Despite the relatively high π_t during the epidemic seasons, thousands of cases may still remain undetected due to the large base of infected individuals, which will increase pressure on the government's epidemic prevention and control efforts. Individuals who are not promptly diagnosed and reported may become potential sources of virus transmission, exacerbating the risk of epidemic spread.

Fig. 7 shows a monotonically increasing relationship between the number of people tested and the reporting probability π_t of influenza. In general, values in periods with a high number of tested (>8000) are about five times higher than those with very low tested (<2000), suggesting that the increase in the tested is a key factor in elevating the reporting probability. The expansion of testing—either in scale or coverage—enhances the likelihood of detecting influenza cases, thereby contributing to a higher reporting probability.

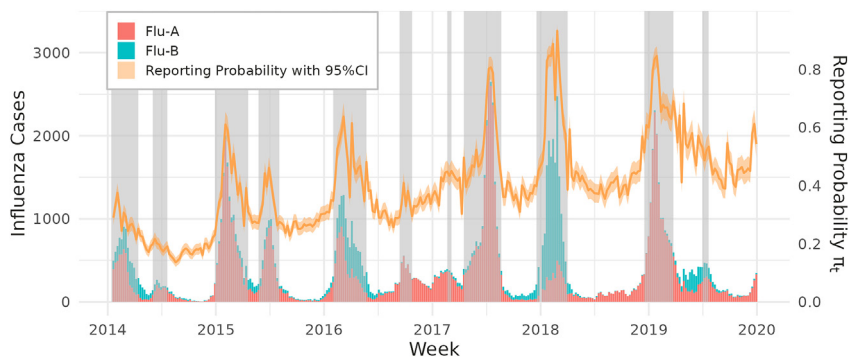


Fig. 6. Reporting probabilities with their 95 % CI (orange) for influenza obtained from the posterior prediction samples estimated by model M_3 , as well as influenza epidemic seasons (grey shaded area) divided based on the baseline threshold of the influenza virus detection positivity rate in Hong Kong of week 1, 2014 to week 52, 2019.

Note: Reporting probability is defined as the probability that an infection is detected and reported in the surveillance system.

Table 4

Summary of the annual mean reporting probabilities for Flu-A and Flu-B in Hong Kong from 2014 to 2019, as well as the mean reporting probabilities during the epidemic season (selecting the longest epidemic season each year).

Year	Type	Epidemic season	Mean of π_t in epidemic season	Mean of π_t in Year
2014	Flu-A	Week3 - week15	22.5 %	16.6 %
	Flu-B		24.1 %	19.8 %
2015	Flu-A	Week1 - week15	35.2 %	26.8 %
	Flu-B		31.7 %	26.6 %
2016	Flu-A	Week6 - week20	40.8 %	30.6 %
	Flu-B		34.4 %	28.3 %
2017	Flu-A	Week17 - week33	48.1 %	37.3 %
	Flu-B		38.7 %	32.2 %
2018	Flu-A	Week52 - week12	63.5 %	42.0 %
	Flu-B		49.5 %	35.9 %
2019	Flu-A	Week51 - week11	61.0 %	46.3 %
	Flu-B		47.0 %	37.8 %

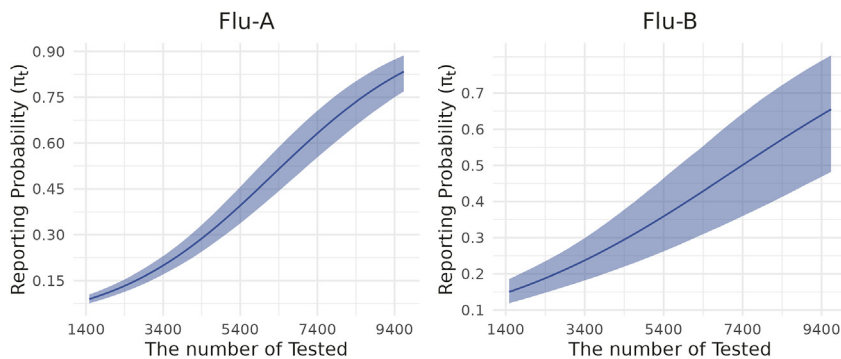


Fig. 7. Posterior mean predicted effect of the tested on the reporting probability of Flu-A and Flu-B, with associated 95 % CI.

Finally, Fig. 8 shows the total observed influenza cases each year, along with the 5 %, 50 %, and 95 % quantiles of the predicted true total. The plot suggests that potentially tens of thousands of cases went unreported each year, posing significant challenges to influenza prevention and control efforts.

Due to delays in the reporting system and the incubation period between influenza infection and symptom onset, there is some lag between the reported incidence and the true transmission rate β_t , as shown in Fig. 9. In the weeks before the epidemic season, β_t increases rapidly and reaches a local maximum, followed by a gradual decrease and stabilization until the start of the next epidemic season.

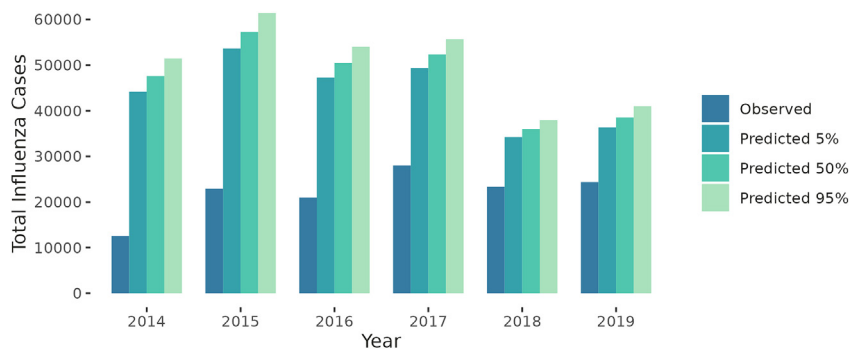


Fig. 8. The recorded total number of influenza case in Hong Kong, as well as the 5 %, 50 %, and 95 % quantiles of the predicted true total number of influenza cases for each year.

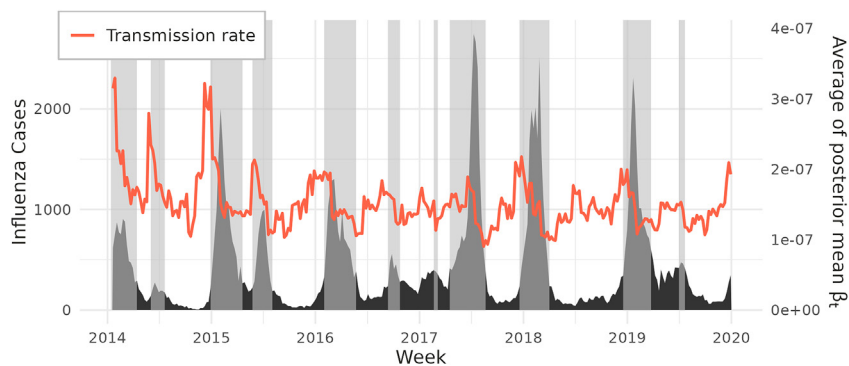


Fig. 9. The posterior mean true weekly transmission rate of influenza in Hong Kong during different epidemic seasons (grey shaded area) from 2014 to 2019.

4. Discussion

Under-reporting is an important problem in disease surveillance to determine the true incidence and burden of disease. Under-reporting surveillance data can lead to biased estimates and potentially wrong decisions. In this study, we used and extended a two-tiered Bayesian model, which consisted of an incidence component model and a case detection probability component model to quantify under-reporting in the region. Compared with existing models, our model has three main innovations. First, we introduced a DT-SIRS model and allowed time-varying transmission rate to capture the dynamic spread of the disease. Second, we incorporated the unknown true disease counts into the compartment model for iterative updating and considered four different transmission rate expressions to improve inference accuracy. Finally, we investigated the relationship between meteorological factors and outbreaks of influenza subtypes in subtropical regions and visualized this relationship through the fluctuations of β_t .

Our study showed that MeanT was negatively associated with influenza and low temperatures led to more influenza cases, which is consistent with previous studies in temperate and subtropical areas (Ma et al., 2022; Ng et al., 2022). Laboratory studies showed that low temperature can promote the spread of influenza by prolonging influenza virus survival, enhancing influenza virus transmissibility and increasing host susceptibility (Lipsitch & Viboud, 2009). In addition, people preferred to stay indoors, which increased the contact rate (Dai et al., 2018). Under the climate background of Hong Kong, we found that Flu-A tends to spread more readily under high RH, which is consistent with a recent study (Ma et al., 2022). Under humid conditions, virus-carrying water droplets are more likely to deposit on surfaces, increasing the chance of contact transmission (Lowen & Palese, 2009). In contrast, more Flu-B outbreaks were observed in dry conditions in the paper. Low RH can allow influenza virus particles to remain airborne longer due to smaller size and slower settling speed (Yang & Marr, 2011). Besides, low RH can also preserve the viability of the influenza virus (Noti et al., 2013).

In our study, both small DTR and large DTR were found to promote influenza transmission. In Macau, influenza risk increased dramatically after exposure to small DTR, regardless of Flu-A or Flu-B (Ng et al., 2022). Additionally, it was observed that more elderly patients were tend to fall ill with influenza in Beijing when there were large temperature variations (Lao et al., 2018). The physiological mechanisms of DTR in influenza were not elucidated, although there were several possible underlying mechanisms (Zhang et al., 2020). Knowledge on the effects of Wv on influenza has been still limited so far. We found that Wv was negatively correlated with Flu-A and positively associated with Flu-B. A large Wv has been shown to facilitate the longer travel of airborne aerosols, but relatively still air might favor the stay and spread of influenza virus in the

same area, leading to more influenza outbreaks in densely populated areas (Ma et al., 2022). Finally, we found that moderate rainfall can increase incidence of Flu-B, while extreme rainfall promotes the transmission of Flu-A, which is consistent with previous results by RH. Extreme rainfall is usually accompanied by high humidity, which increases the chances for contact transmission of the influenza virus.

Additionally, our results indicated that there were significant differences in the reporting probability of two influenza subtypes during different epidemic periods in Hong Kong. Throughout the study period, the annual average reporting probability for both Flu-A and Flu-B showed an increasing trend over time. Within individual years, the average reporting probability was higher during epidemic periods than during non-epidemic periods. This reflected both the government's rational allocation of public health resources and the improvement of people's self-protection awareness. As the results shown in Figs. 7–8, the relative difference between the predicted true total and the observed counts gradually decreases as the number of tested increases. This will allow us to assess the true scale of pandemic more accurately, which in turn will help policy makers formulate more effective prevention and control strategies to protect public health.

In recent years, various approaches have been employed to correct the under-reporting in influenza surveillance data, ranging from simple multipliers to more complex mathematical and statistical models (Reed et al., 2015). Our model significantly simplifies the assumptions that we don't need to require all individuals to be susceptible. Compared with the Poisson-logistic model, our model offers a more accurate derivation of the true disease counts when simulating over longer periods of time. Moreover, our model allows for the estimation of the predictive distribution of reporting probability and the true transmission rate, thus providing a better quantification of the uncertainty associated with under-reporting and transmission.

In addition, our model used true disease counts in the recursive equations, which allowed the calculation of the true transmission rate β_t and influenza incidence without under-estimation. And we could see a sharp rise in β_t in the first few weeks before the epidemic season, then followed by a rapid increase in influenza cases. This lagging phenomenon provides new insights for early warning of influenza. The transmission trend of influenza can be assessed more accurately through timely investigating the fluctuations of β_t . The calculation of β_t is closely tied to meteorological factors. Currently, short-term forecasts based on meteorological factors are currently widely applied (Ma, Guo, et al., 2022), and some of the meteorological data we use are lagged as shown in Section 3.1, allowing us to make reasonable predictions about β_t in the next two or three weeks.

Our model has several practical applications. First, by quantifying the extent of under-reporting, health authorities can better allocate diagnostic resources and strengthen monitoring and screening of influenza cases in critical periods. Second, the integration of meteorological data allows the system to serve as a season-aware early warning tool. By detecting the trend of rapid increase of β_t in advance and issuing warning signals, corresponding prevention and control measures can be taken.

Although the findings of this study are relatively robust, several limitations should be acknowledged. First, our model does not account for spatial heterogeneity. Regional differences may affect the generalizability of the results. Second, due to the unavailability of data, we were unable to differentiate between subtypes of Influenza A and B. This may mask important differences in transmission dynamics and reporting probabilities among subtypes, thereby affecting the precision of our estimates. Third, the inclusion of covariates in the model remains limited. On one hand, only meteorological factors were considered as covariates influencing transmission rate, which may not fully capture the multifaceted drivers of influenza spread. On the other hand, individuals with recent infections or vaccinations—who tend to exhibit milder symptoms and are less likely to seek medical care—are difficult to account for using current detection-related covariates, potentially leading to under-reporting. Moreover, as our model relies on influenza surveillance data, its applicability may be limited in regions with sparse or incomplete testing data.

There are some generalizations to our research. An extension of the proposed model is its application to spatial domains to determine potentially under-reported areas. Another important area for future research is the inclusion of covariates that influence disease transmission or detection, such as socioeconomic factors or human activities, which would significantly improve the description and inference of the dynamics of infectious disease.

5. Conclusion

Our method provides a straightforward approach to better characterize the under-reported of influenza that could be efficiently and rapidly performed only using routine surveillance data and meteorological variables. In this study, we combined the Poisson-logistic model with the DT-SIRS model through an extended incidence component to create a new model and considered four different transmission rate expressions to capture seasonal effects, and further estimated all parameters in the Bayesian framework. The results of DIC and RMSE indicated that our model fits the data well. Compared with the Poisson-logistic model, the new model made the derivation of true disease counts and true transmission rates more reasonable by including covariates that impact disease detection and transmission, resulting in more convincing estimates of under-reporting.

CRedit authorship contribution statement

Peiji Li: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Mengmeng Dai:** Validation, Investigation. **Yayi Wang:** Writing – review & editing, Formal analysis. **Yingbo Liu:** Supervision, Methodology.

Ethics approval and consent to participate

The influenza data analyzed in the study were collected as part of routine surveillance program and do not involve personal information, so it is exempt from ethical approval.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2025.05.002>.

References

- Bailey, T. C., Carvalho, M. S., Lapa, T. M., Souza, W. V., & Brewer, M. J. (2005). Modeling of under-detection of cases in disease surveillance. *Annals of Epidemiology*, 15(5), 335–343. <https://doi.org/10.1016/j.annepidem.2004.09.013>
- Bucyibaruta, G., Dean, C. B., & Torabi, M. (2023). A discrete-time susceptible-infectious-recovered-susceptible model for the analysis of influenza data. *Infectious Disease Modelling*, 8(2), 471–483. <https://doi.org/10.1016/j.idm.2023.04.008>
- Centre for Health Protection. Flu express Retrieved from. <https://www.chp.gov.hk/en/resources/29/304.html>. (Accessed 1 November 2023).
- Centre for Health Protection/Department of Health of the Government of the Hong Kong Special Administrative Region. (2023). *General out-patient clinics/private medical practitioner clinics*. <https://www.chp.gov.hk/en/static/24015.html> (Accessed 1 November 2023).
- Cohen, C., Kleynhans, J., Moyes, J., McMorrow, M. L., Treurnicht, F. K., Hellferscee, O., Mathunjwa, A., Gottberg, A. von, Wolter, N., Martinson, N. A., Kahn, K., Lebina, L., Mothlaoleng, K., Wafawanaka, F., Gómez-Olivé, F. X., Mkhencle, T., Mathee, A., Piketh, S., Language, B., ... Gómez-Olivé, F. X. (2021). Asymptomatic transmission and high community burden of seasonal influenza in an urban and a rural community in South Africa, 2017–18 (phirst): A population cohort study. *Lancet Global Health*, 9(6), e863–e874. [https://doi.org/10.1016/S2214-109X\(21\)00141-8](https://doi.org/10.1016/S2214-109X(21)00141-8)
- Corberán-Vallet, A., & Santonja, F. J. (2014). A Bayesian SIRS model for the analysis of respiratory syncytial virus in the region of Valencia, Spain. *Biometrical Journal*, 56(5), 808–818. <https://doi.org/10.1002/bimj.201300194>
- Dai, Q., Ma, W., Huang, H., Xu, K., Qi, X., Yu, H., Deng, F., Bao, C., & Huo, X. (2018). The effect of ambient temperature on the activity of influenza and influenza like illness in Jiangsu Province, China. *Science of the Total Environment*, 645, 684–691. <https://doi.org/10.1016/j.scitotenv.2018.07.065>
- de Oliveira, G. L., Loschi, R. H., & Assuncao, R. M. (2017). A random-censoring Poisson model for underreported data. *Statistics in Medicine*, 36(30), 4873–4892. <https://doi.org/10.1002/sim.7456>
- de Valpine, P., Turek, D., Paciorek, C. J., Anderson-Bergman, C., Lang, D. T., & Bodik, R. (2017). Programming with models: Writing statistical algorithms for general model structures with NIMBLE. *Journal of Computational & Graphical Statistics*, 26(2), 403–413. <https://doi.org/10.1080/10618600.2016.1172487>
- Domenech De Cellès, M., Arduin, H., Lévy-Bruhl, D., Georges, S., Souty, C., Guillemot, D., Watier, L., & Opatowski, L. (2019). Unraveling the seasonal epidemiology of pneumococcus. *Proceedings of the National Academy of Sciences*, 116(5), 1802–1807. <https://doi.org/10.1073/pnas.1812388116>
- Fielding, J. E., Kelly, H. A., & Glass, K. (2015). Transmission of the first influenza A(H1N1) pdm09 pandemic wave in Australia was driven by undetected infections: Pandemic response implications. *PLoS One*, 10(12), Article e0144331. <https://doi.org/10.1371/journal.pone.0144331>
- Gasparrini, A. (2011). Distributed lag linear and non-linear models in R: The package dlnm. *Journal of Statistical Software*, 43(8), 1–20. <https://doi.org/10.18637/jss.v043.i08>
- He, D., Dushoff, J., Eftimie, R., & Earn, D. J. D. (2013). Patterns of spread of influenza A in Canada. *Proceedings of the Royal Society B: Biological Sciences*, 280(1770), Article 20131174. <https://doi.org/10.1098/rspb.2013.1174>
- Hong Kong Observatory. Automatic Weather Observations daily Extract. Retrieved from <https://www.hko.gov.hk/en/cis/awsdailyextract.htm?stn=HKA>. (Accessed 1 November 2023).
- Kim, S., Lee, J., & Jung, E. (2017). Mathematical model of transmission dynamics and optimal control strategies for 2009 A/H1N1 influenza in the Republic of Korea. *Journal of Theoretical Biology*, 412, 74–85. <https://doi.org/10.1016/j.jtbi.2016.09.025>
- Kupek, E. (2021). How many more? Under-Reporting of the COVID-19 deaths in Brazil in 2020. *Tropical Medicine and International Health*, 26(9), 1019–1028. <https://doi.org/10.1111/tmi.13628>
- Lao, J. H., Liu, Z. D., Liu, Y. Y., Zhang, J., & Jiang, B. F. (2018). [Influence of diurnal temperature range on influenza incidence in the elderly]. *Zhonghua liu xing bing xue za zhi = Zhonghua liuxingbingxue zazhi*, 39(11), 1454–1458. <https://doi.org/10.3760/cma.j.issn.0254-6450.2018.11.007>
- Lau, L. L. H., Cowling, B. J., Fang, V. J., Chan, K., Lau, E. H. Y., Lipsitch, M., Cheng, C. K. Y., Houck, P. M., Uyeki, T. M., Peiris, J. S. M., & Leung, G. M. (2010). Viral shedding and clinical illness in naturally acquired influenza virus infections. *The Journal of Infectious Diseases*, 201(10), 1509–1516. <https://doi.org/10.1086/652241>
- Lawson, A. B., & Song, H.-R. (2010). 36 Bayesian hierarchical modeling of the dynamics of spatio-temporal influenza season outbreaks. *Spatial and Spatio-Temporal Epidemiology*, 1(2–3), 187–195. <https://doi.org/10.1016/j.sste.2010.03.001>
- Lipsitch, M., & Viboud, C. (2009). Influenza seasonality: Lifting the fog. *Proceedings of the National Academy of Sciences*, 106(10), 3645–3646. <https://doi.org/10.1073/pnas.0900933106>
- Liu, T., Kang, M., Zhang, B., Xiao, J., Lin, H., Zhao, Y., Huang, Z., Wang, X., Zhang, Y., He, J., & Ma, W. (2018). Independent and interactive effects of ambient temperature and absolute humidity on the risks of avian influenza A(H7N9) infection in China. *Science of the Total Environment*, 619, 1358–1365. <https://doi.org/10.1016/j.scitotenv.2017.11.226>
- Lope, D. J., & Demiran, H. (2022). Spatiotemporal Bayesian estimation of the number of under-reported COVID-19 cases in Victoria Australia. *PeerJ*, 10, Article e14184. <https://doi.org/10.7717/peerj.14184>
- Lowen, A., & Palese, P. (2009). Transmission of influenza virus in temperate zones is predominantly by aerosol, in the tropics by contact: A hypothesis. *PLoS Currents*, 1, Article RRN1002.
- Ma, P., Tang, X., Zhang, L., Wang, X., Wang, W., Zhang, X., Wang, S., & Zhou, N. (2022). Influenza A and B outbreaks differed in their associations with climate conditions in Shenzhen, China. *International Journal of Biometeorology*, 66(1), 163–173. <https://doi.org/10.1007/s00484-021-02204-y>
- Ma, Z., Guo, G., Cai, M., Chen, X., Li, W., & Zhang, L. (2022). A combined linear-nonlinear short-term rainfall forecast method using GNSS-derived PWV. *Atmosphere*, 13(9), 1381. <https://doi.org/10.3390/atmos13091381>

- Montgomery, M. P., Morris, S. E., Rolfes, M. A., Kittikraisak, W., Samuels, A. M., Biggerstaff, M., Davis, W. W., Reed, C., & Olsen, S. J. (2024). The role of asymptomatic infections in influenza transmission: What do we really know. *The Lancet Infectious Diseases*, 24(6), e394–e404. [https://doi.org/10.1016/S1473-3099\(23\)00619-9](https://doi.org/10.1016/S1473-3099(23)00619-9)
- Ng, H., Li, Y., Zhang, T., Lu, Y., Wong, C., Ni, J., & Zhao, Q. (2022). Association between multiple meteorological variables and seasonal influenza A and B virus transmission in Macau. *Heliyon*, 8(11), Article e11820. <https://doi.org/10.1016/j.heliyon.2022.e11820>
- Noti, J. D., Blachere, F. M., McMillen, C. M., Lindsley, W. G., Kashon, M. L., Slaughter, D. R., & Beezhold, D. H. (2013). High humidity leads to loss of infectious influenza virus from simulated coughs. *PLoS One*, 8(2), Article e57485. <https://doi.org/10.1371/journal.pone.0057485>
- Reed, C., Chaves, S. S., Kirley, P. D., Emerson, R., Aragon, D., Hancock, E. B., Butler, L., Baumbach, J., Hollick, G., Bennett, N. M., Laidler, M. R., Thomas, A., Meltzer, M. I., & Finelli, L. (2015). Estimating influenza disease burden from population-based surveillance data in the United States. *PLoS One*, 10(3), Article e0118369. <https://doi.org/10.1371/journal.pone.0118369>
- Stoner, O., Economou, T., & Drummond Marques Da Silva, G. (2019). A hierarchical framework for correcting under-reporting in count data. *Journal of the American Statistical Association*, 114(528), 1481–1492. <https://doi.org/10.1080/01621459.2019.1573732>
- Wang, L., Liu, Z., Guo, C., Li, Y., & Zhang, X. (2021). New global dynamical results and application of several SVEIS epidemic models with temporary immunity. *Applied Mathematics and Computation*, 390, Article 125648. <https://doi.org/10.1016/j.amc.2020.125648>
- Winkelmann, R., & Zimmermann, K. F. (1993). *Poisson-logistic regression*. München: Volkswirtschaftl. Fak., Ludwig-Maximilians-Univ.
- World Health Organization. (2018). Up to 650 000 people die of respiratory diseases linked to seasonal flu each year. Retrieved from <https://www.who.int/news/item/13-12-2017-up-to-650-000-people-die-of-respiratory-diseases-linked-to-seasonal-flu-each-year>. (Accessed 1 March 2024).
- Yaari, R., Katriel, G., Huppert, A., Axelsen, J. B., & Stone, L. (2013). Modelling seasonal influenza: The role of weather and punctuated antigenic drift. *Journal of The Royal Society Interface*, 10(84), Article 20130298. <https://doi.org/10.1098/rsif.2013.0298>
- Yang, W., & Marr, L. C. (2011). Dynamics of airborne influenza A viruses indoors and dependence on humidity. *PLoS One*, 6(6), Article e21481. <https://doi.org/10.1371/journal.pone.0021481>
- Zhang, Y., Ye, C., Yu, J., Zhu, W., Wang, Y., Li, Z., Xu, Z., Cheng, J., Wang, N., Hao, L., & Hu, W. (2020). The complex associations of climate variability with seasonal influenza A and B virus transmission in subtropical Shanghai, China. *Science of the Total Environment*, 701, Article 134607. <https://doi.org/10.1016/j.scitotenv.2019.134607>